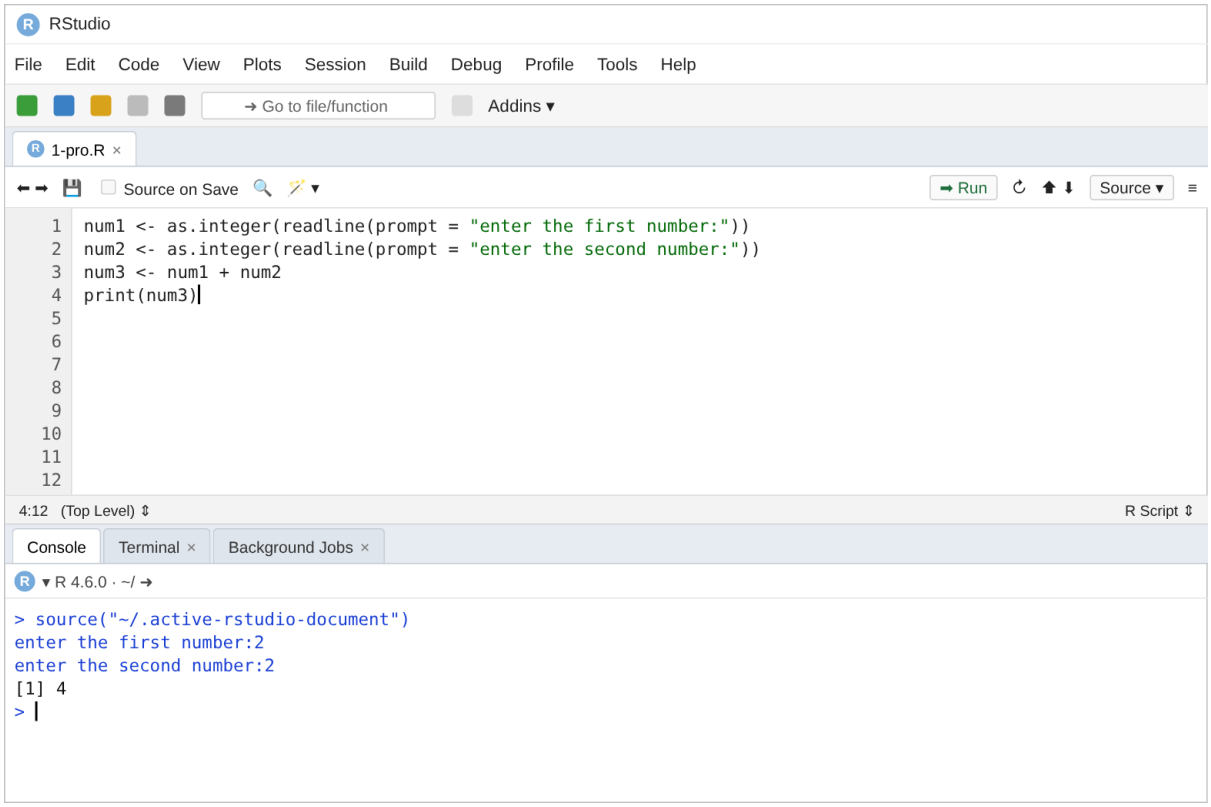
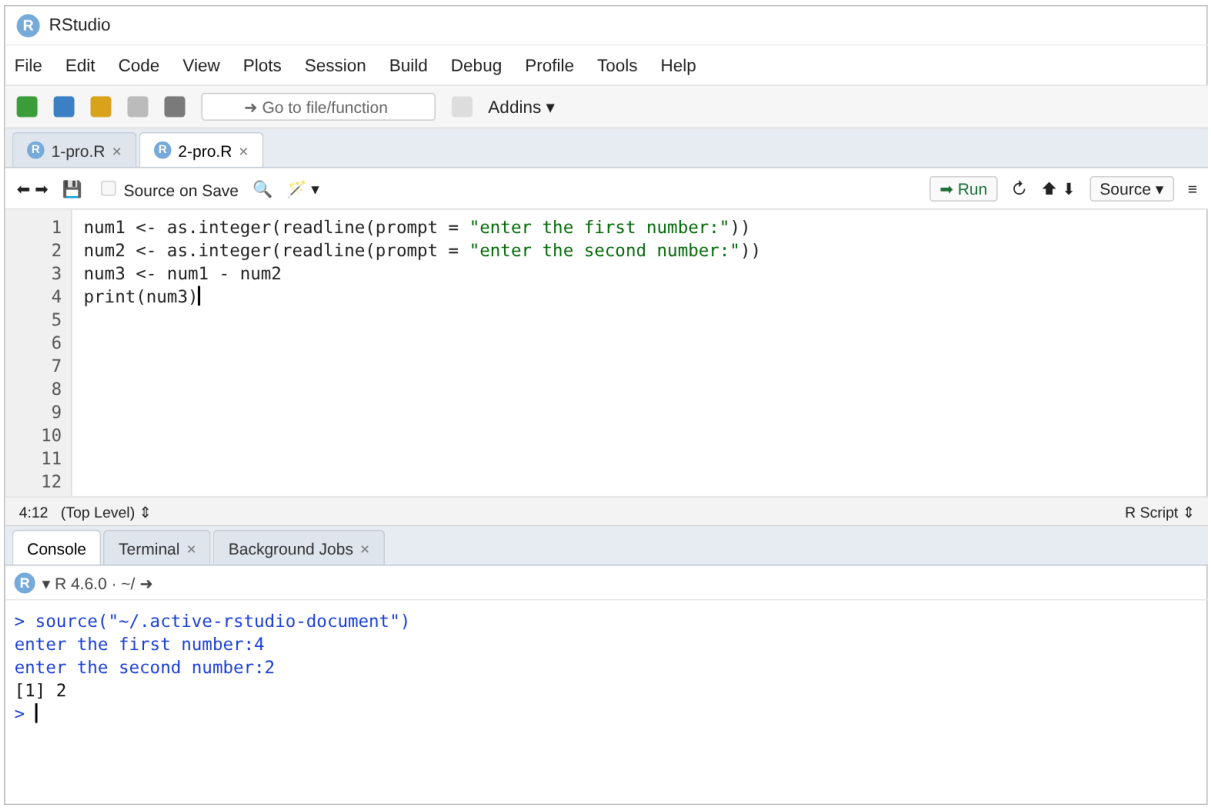
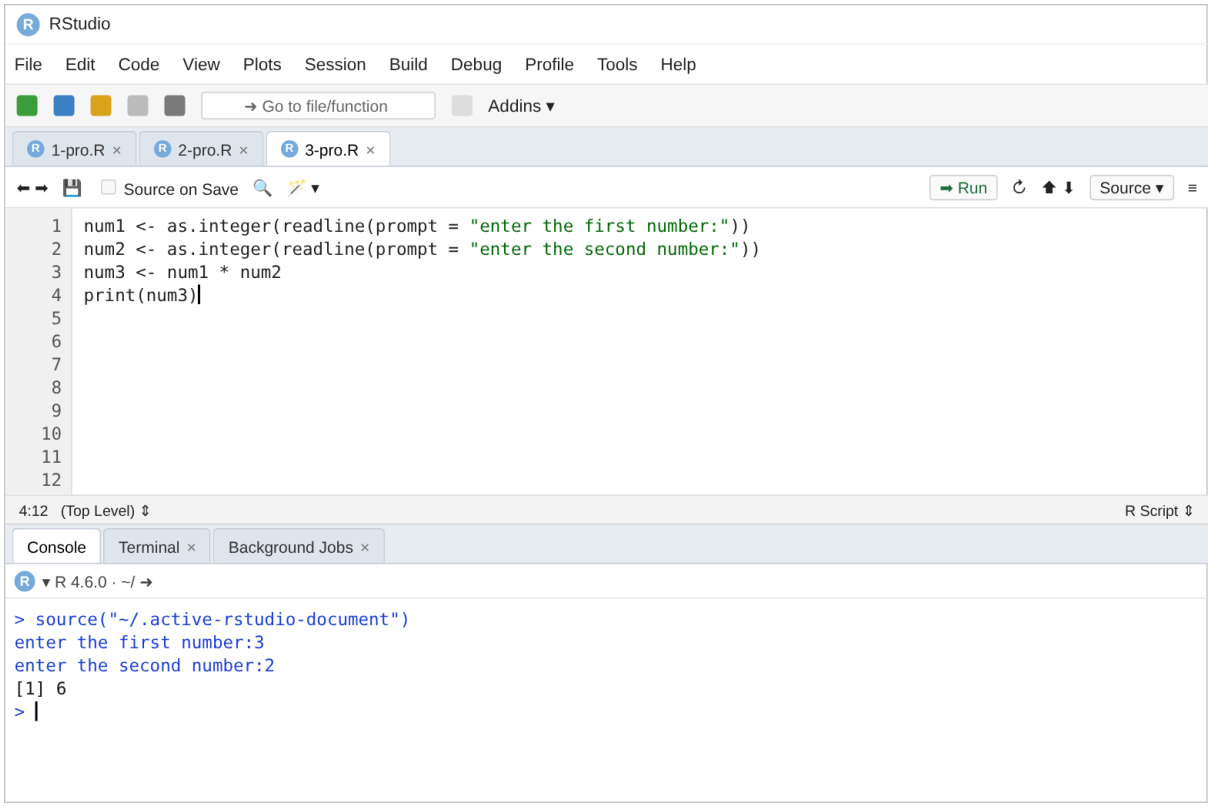
EXP-1


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
Source on Save
Run
Source
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 + num2
print(num3)
4:12 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
enter the first number:2
enter the second number:2
[1] 4
>

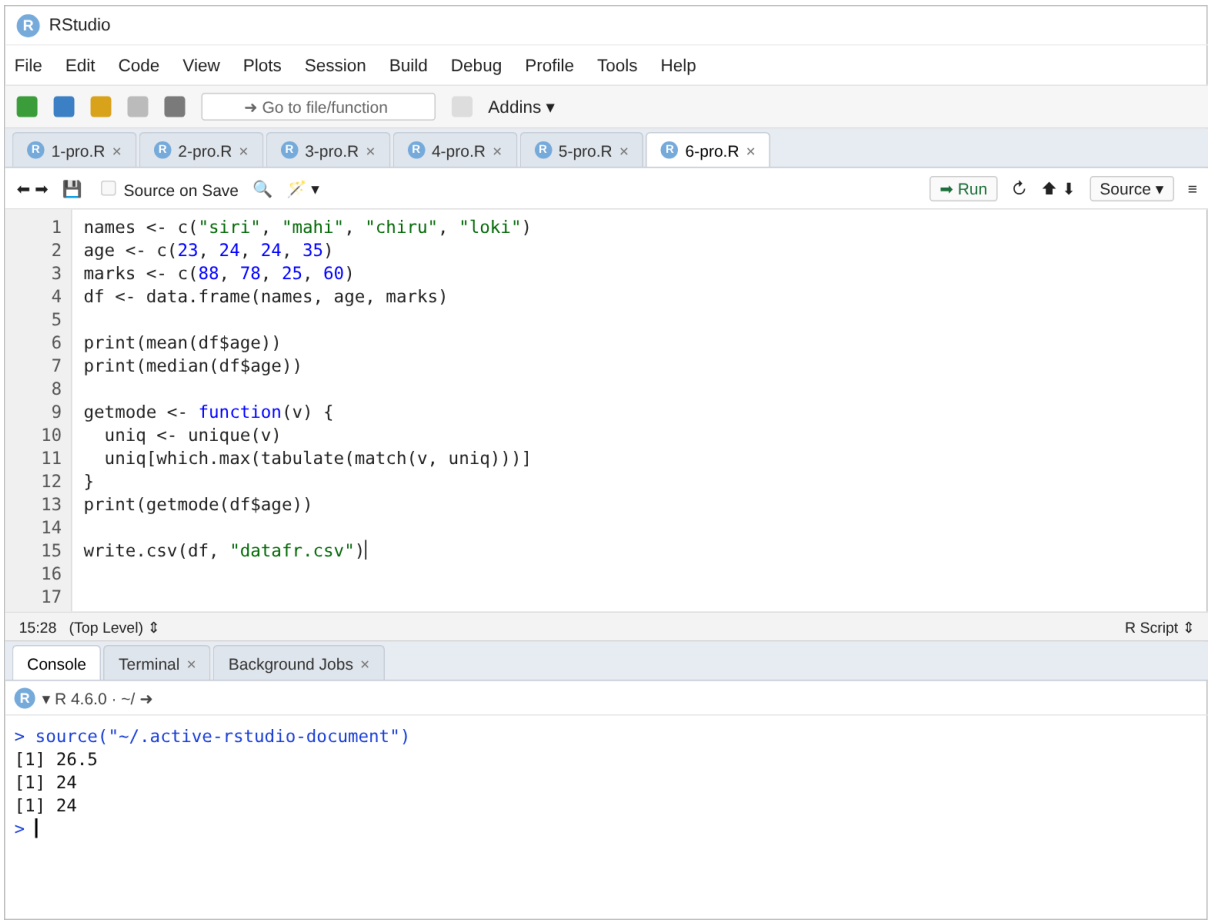
EXP-2


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
Source on Save
Run
Source
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 - num2
print(num3)
4:12 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
enter the first number:4
enter the second number:2
[1] 2
>

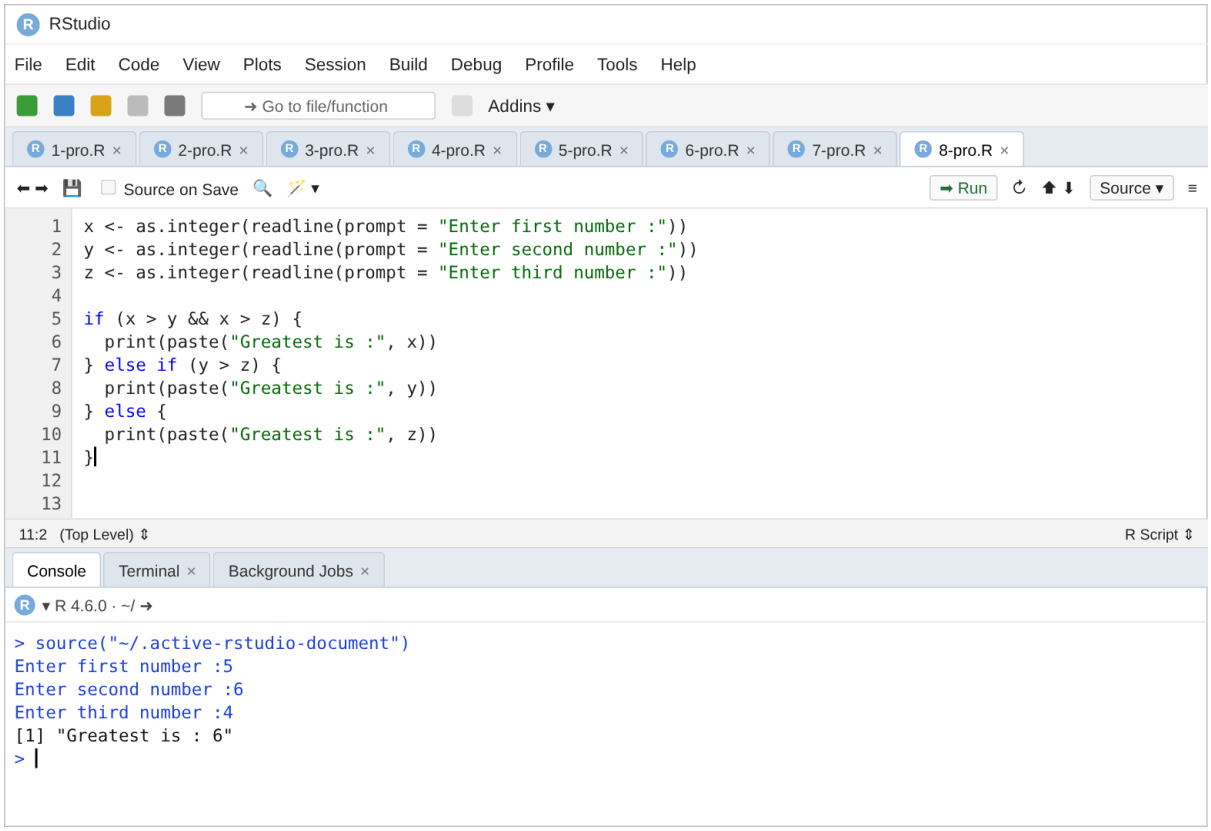
EXP-3


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
3-pro.R
Source on Save
Run
Source
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 * num2
print(num3)
4:12 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
enter the first number:3
enter the second number:2
[1] 6
>

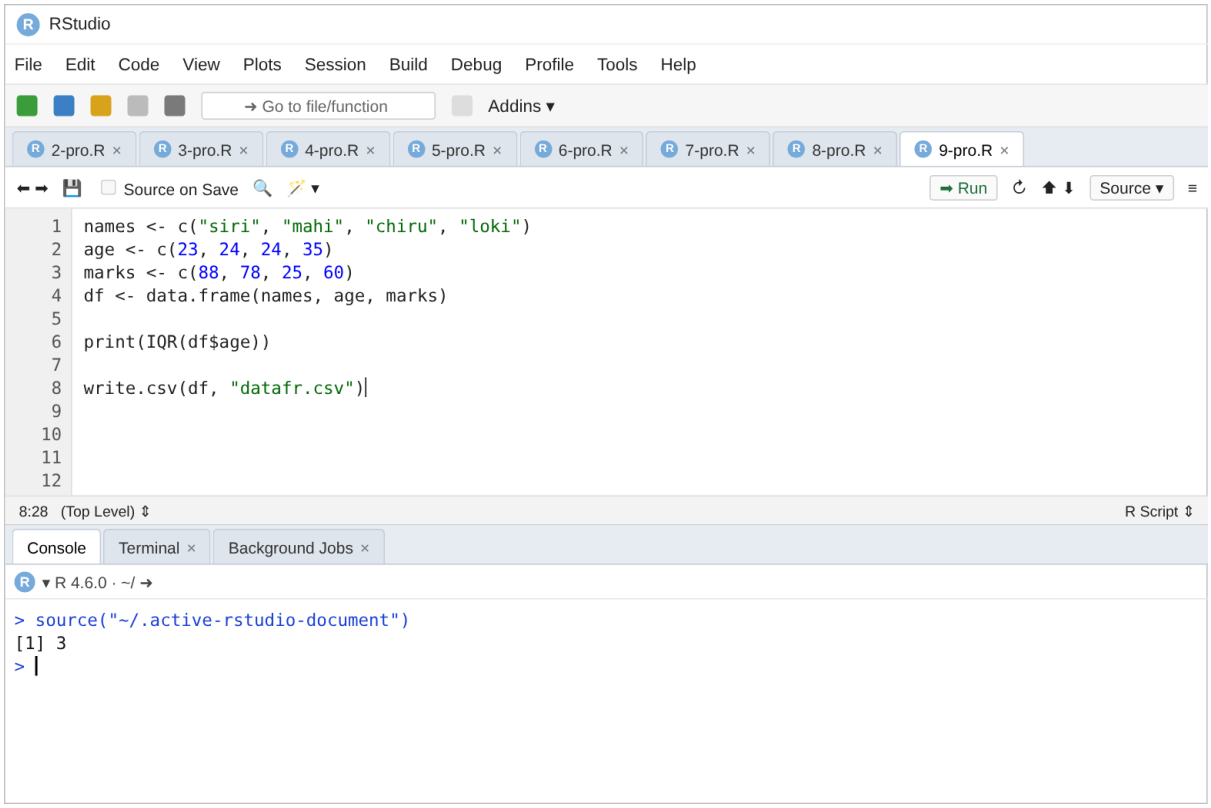
EXP-4


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
3-pro.R
4-pro.R
5-pro.R
6-pro.R
Source on Save
Run
Source
names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)
print(mean(df$age))
print(median(df$age))
getmode <- function(v) {
uniq <- unique(v)
uniq[which.max(tabulate(match(v, uniq)))]
}
print(getmode(df$age))
write.csv(df, "datafr.csv")
15:28 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 26.5
[1] 24
[1] 24
>

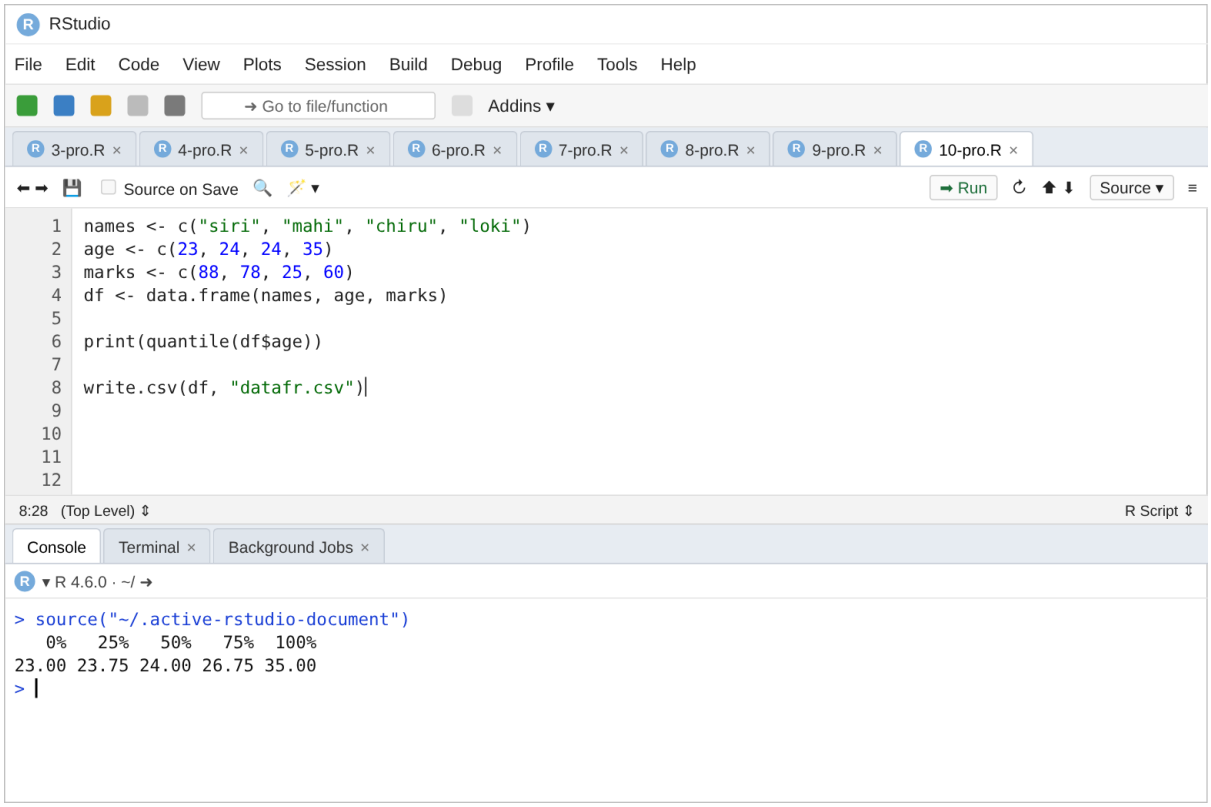
EXP-5


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
3-pro.R
4-pro.R
5-pro.R
6-pro.R
7-pro.R
8-pro.R
Source on Save
Run
Source
x <- as.integer(readline(prompt = "Enter first number :"))
y <- as.integer(readline(prompt = "Enter second number :"))
z <- as.integer(readline(prompt = "Enter third number :"))
if (x > y && x > z) {
print(paste("Greatest is :", x))
} else if (y > z) {
print(paste("Greatest is :", y))
} else {
print(paste("Greatest is :", z))
}
11:2 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
Enter first number :5
Enter second number :6
Enter third number :4
[1] "Greatest is : 6"
>

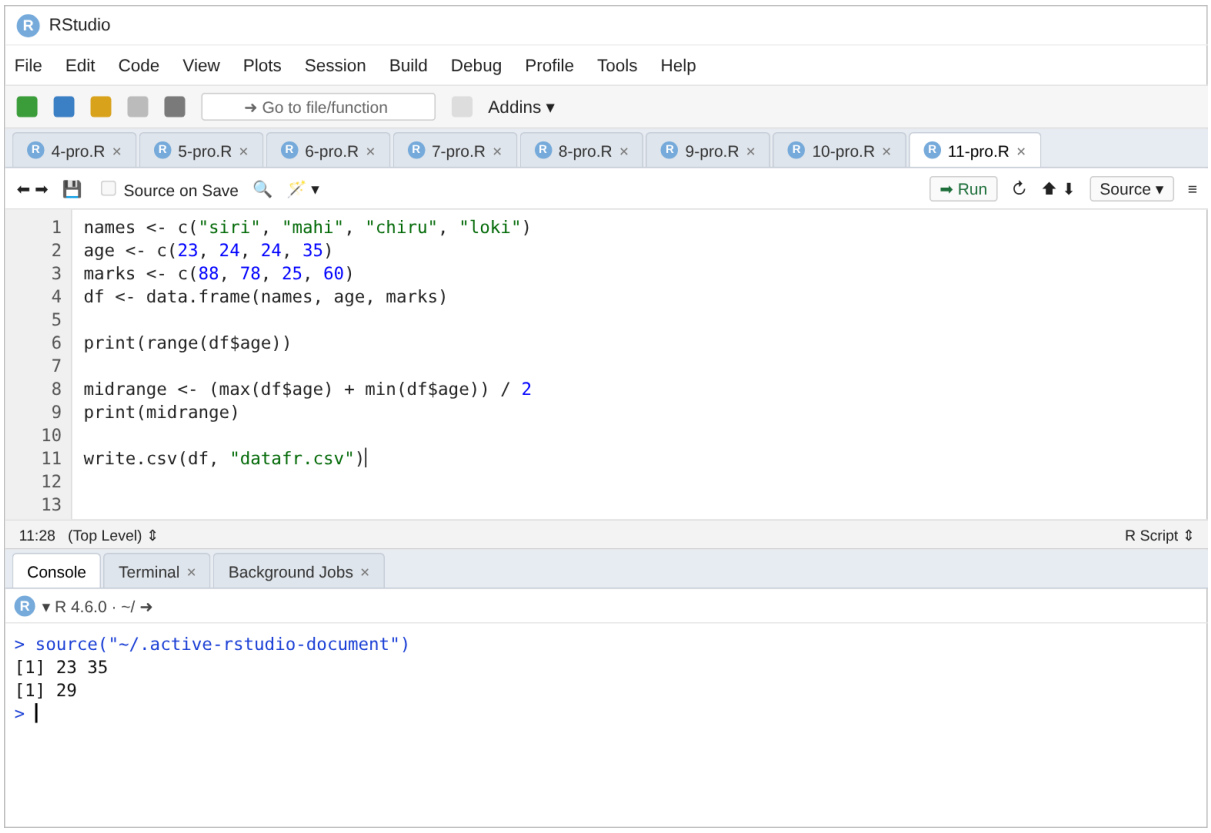
EXP-6


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
2-pro.R
3-pro.R
4-pro.R
5-pro.R
6-pro.R
7-pro.R
8-pro.R
9-pro.R
Source on Save
Run
Source
names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)
print(IQR(df$age))
write.csv(df, "datafr.csv")
8:28 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 3
>

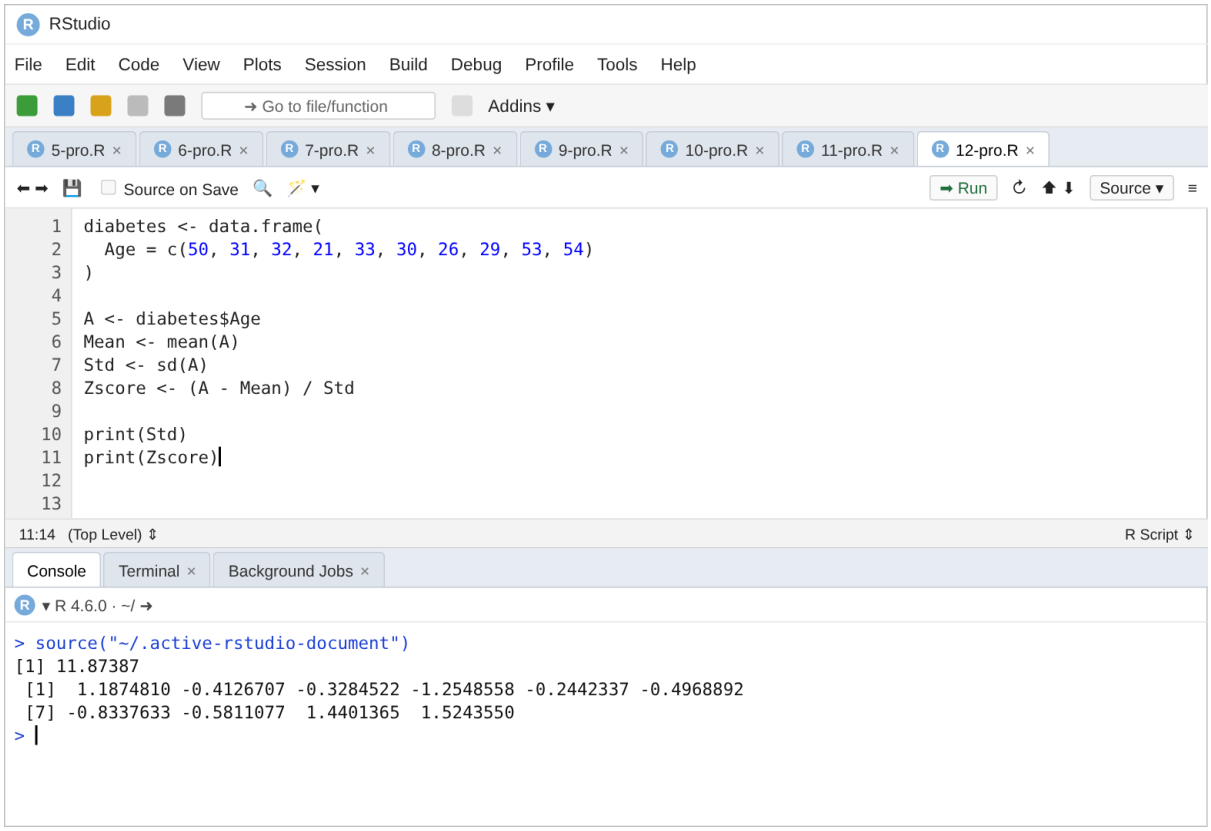
EXP-7


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
3-pro.R
4-pro.R
5-pro.R
6-pro.R
7-pro.R
8-pro.R
9-pro.R
10-pro.R
Source on Save
Run
Source
1 names <- c("siri", "mahi", "chiru", "loki")
2 age <- c(23, 24, 24, 35)
3 marks <- c(88, 78, 25, 60)
4 df <- data.frame(names, age, marks)
5
6 print(quantile(df$age))
7
8 write.csv(df, "datafr.csv")
9
10
11
12
8:28 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
0% 25% 50% 75% 100%
23.00 23.75 24.00 26.75 35.00
>

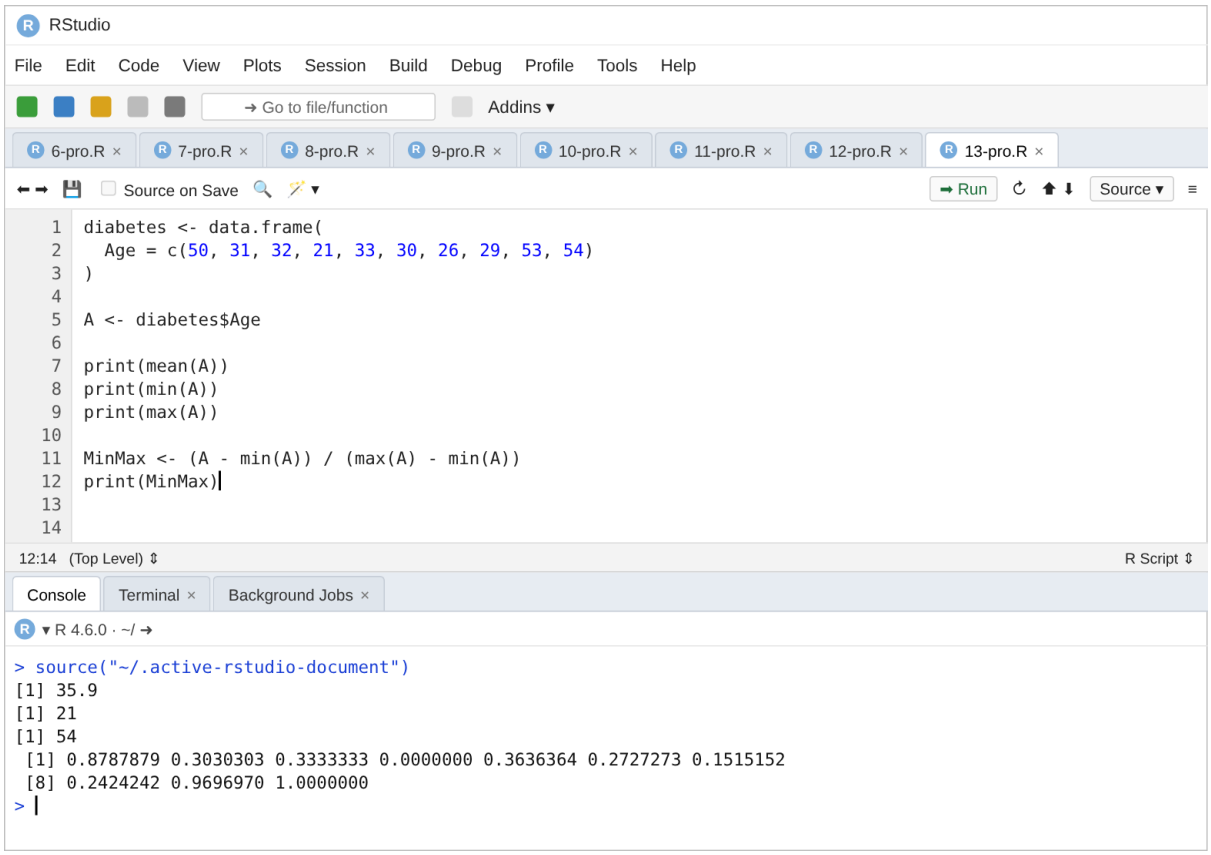
EXP-8


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
4-pro.R
5-pro.R
6-pro.R
7-pro.R
8-pro.R
9-pro.R
10-pro.R
11-pro.R
Source on Save
Run
Source
1 names <- c("siri", "mahi", "chiru", "loki")
2 age <- c(23, 24, 24, 35)
3 marks <- c(88, 78, 25, 60)
4 df <- data.frame(names, age, marks)
5
6 print(range(df$age))
7
8 midrange <- (max(df$age) + min(df$age)) / 2
9 print(midrange)
10
11 write.csv(df, "datafr.csv")
12
13
11:28 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 23 35
[1] 29
>

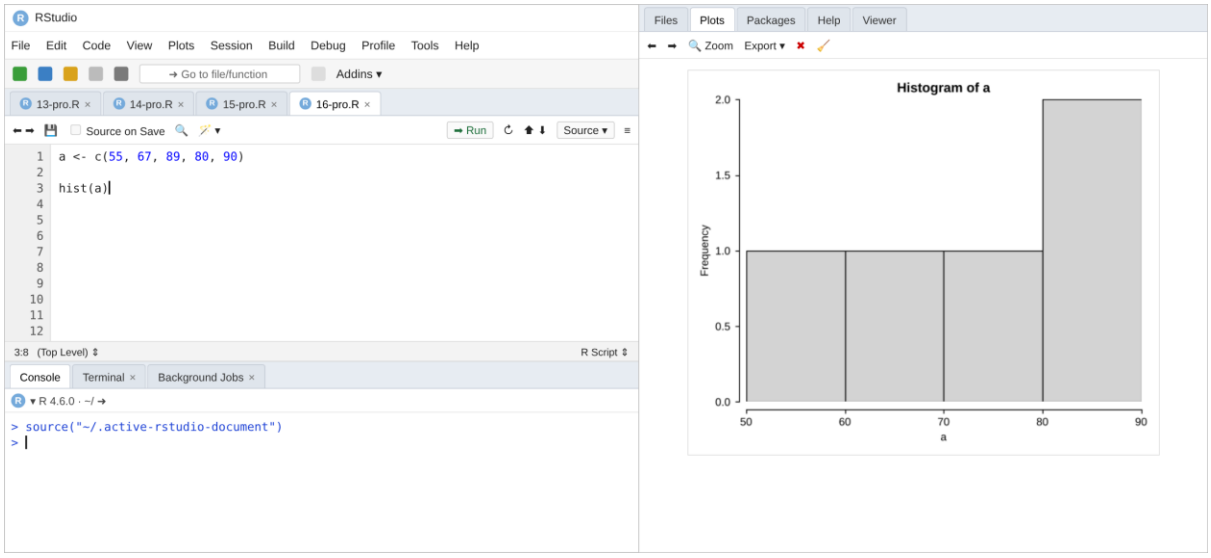
EXP-9


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
5-pro.R
6-pro.R
7-pro.R
8-pro.R
9-pro.R
10-pro.R
11-pro.R
12-pro.R
Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
3)
4
5 A <- diabetes$Age
6 Mean <- mean(A)
7 Std <- sd(A)
8 Zscore <- (A - Mean) / Std
9
10 print(Std)
11 print(Zscore)
12
13
11:14 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 11.87387
[1] 1.1874810 -0.4126707 -0.3284522 -1.2548558 -0.2442337 -0.4968892
[7] -0.8337633 -0.5811077 1.4401365 1.5243550
>

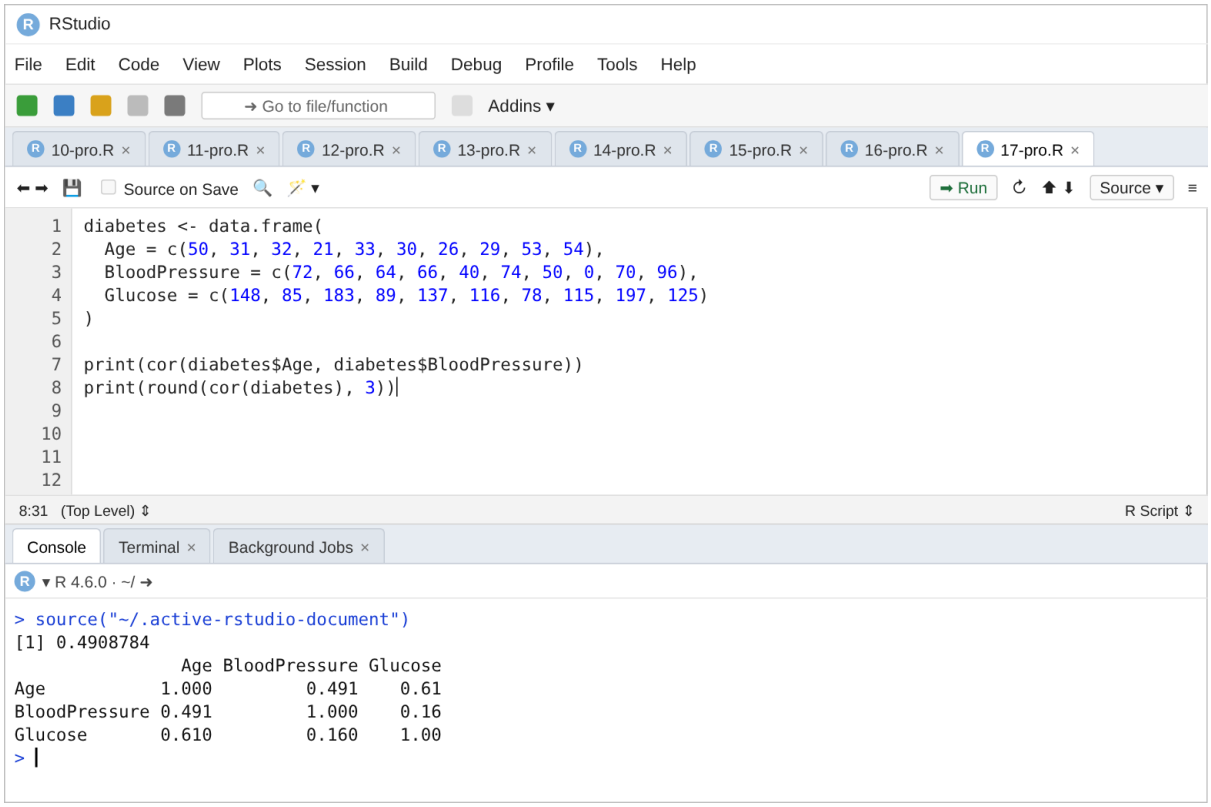
EXP-10


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
6-pro.R
7-pro.R
8-pro.R
9-pro.R
10-pro.R
11-pro.R
12-pro.R
13-pro.R
Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
3)
4
5 A <- diabetes$Age
6
7 print(mean(A))
8 print(min(A))
9 print(max(A))
10
11 MinMax <- (A - min(A)) / (max(A) - min(A))
12 print(MinMax)
13
14
12:14 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 35.9
[1] 21
[1] 54
[1] 0.8787879 0.3030303 0.3333333 0.0000000 0.3636364 0.2727273 0.1515152
[8] 0.2424242 0.9696970 1.0000000
>

EXP-11


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
13-pro.R
14-pro.R
15-pro.R
16-pro.R
Source on Save
Run
Source
1 a <- c(55, 67, 89, 80, 90)
3 hist(a)
3:8 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
>
Files
Plots
Packages
Help
Viewer
Zoom
Export
Histogram of a
Frequency
2.0
1.5
1.0
0.5
0.0
50
60
70
80
90
a

EXP-12


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
10-pro.R
11-pro.R
12-pro.R
13-pro.R
14-pro.R
15-pro.R
16-pro.R
17-pro.R
Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
3 BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
4 Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
5)
7 print(cor(diabetes$Age, diabetes$BloodPressure))
8 print(round(cor(diabetes), 3))
8:31 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
[1] 0.4908784
Age BloodPressure Glucose
Age 1.000 0.491 0.61
BloodPressure 0.491 1.000 0.16
Glucose 0.610 0.160 1.00
>

EXP-13

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

16-pro.R | 17-pro.R | 18-pro.R | 19-pro.R

Source on Save | Run | Source

```
diabetes <- data.frame(
  Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
  BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
  Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
)

Relation <- lm(BloodPressure ~ Age, data = diabetes)
print(Relation)

plot(diabetes$Age, diabetes$BloodPressure,
     col = "darkgreen",
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "BloodPressure")
abline(Relation, col = "red")
```

14:30 (Top Level) | R Script

Console | Terminal | Background Jobs

R 4.6.0 · ~/

```
> source("~/.active-rstudio-document")

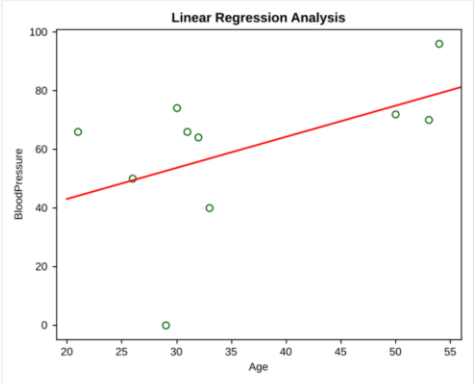
Call:
lm(formula = BloodPressure ~ Age, data = diabetes)

Coefficients:
(Intercept)          Age
     21.724        1.061

>
```

Files | Plots | Packages | Help | Viewer

Zoom | Export



EXP-14

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins ▾

13-pro.R × | 14-pro.R × | 15-pro.R × | 16-pro.R × | 17-pro.R × | 18-pro.R × | 19-pro.R × | 20-pro.R ×

Source on Save | Run | Source ▾

```
1  diabetes <- data.frame(
2    Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
3    BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
4    Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
5  )
6
7  input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
8  model <- lm(Age ~ BloodPressure + Glucose, data = input)
9  print(model)
10
11 A <- coef(model)[[1]]
12 xBloodPressure <- coef(model)[[2]]
13 yGlucose <- coef(model)[[3]]
14
15 print(A)
16 print(xBloodPressure)
17 print(yGlucose)
18
19 y <- A + xBloodPressure * 80 + yGlucose * 120
20 print(y)
21
22
```

20:9 (Top Level) ⇕ R Script ⇕

Console | Terminal × | Background Jobs ×

▾ R 4.6.0 · ~/ ➜

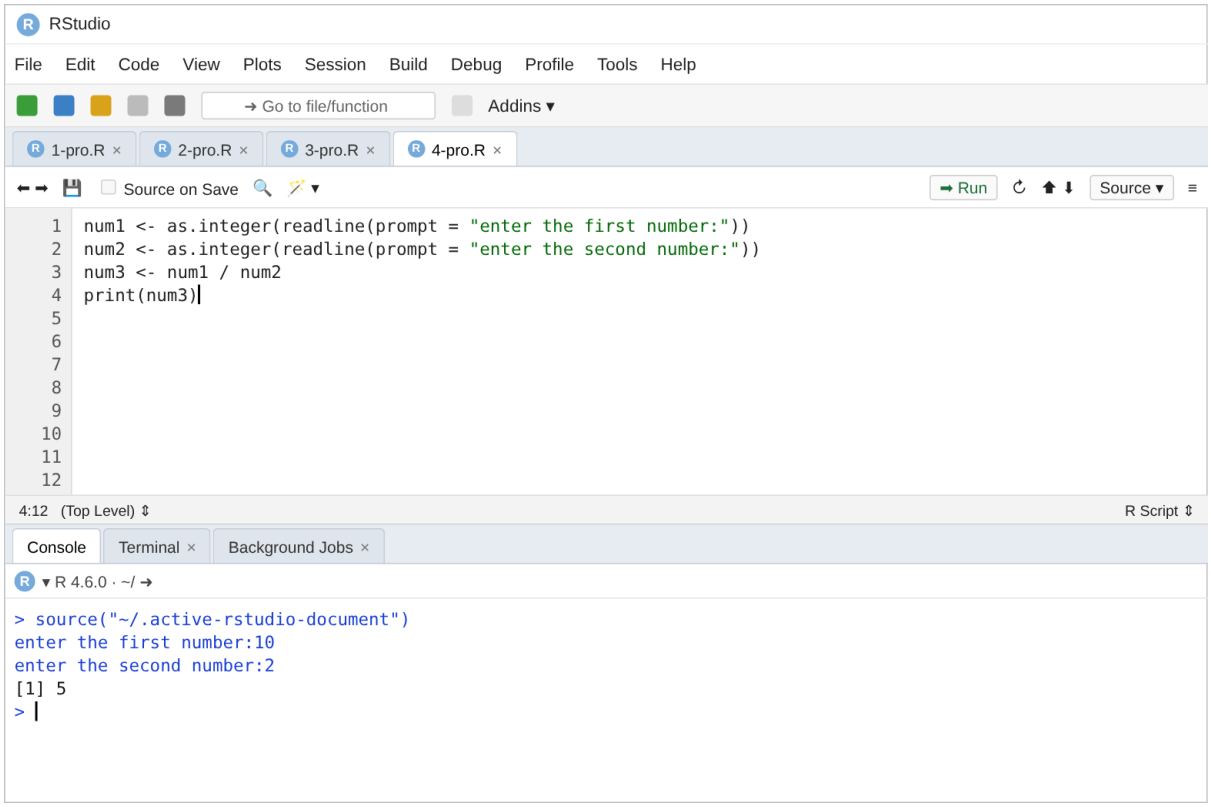
```
> source("~/.active-rstudio-document")

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = input)

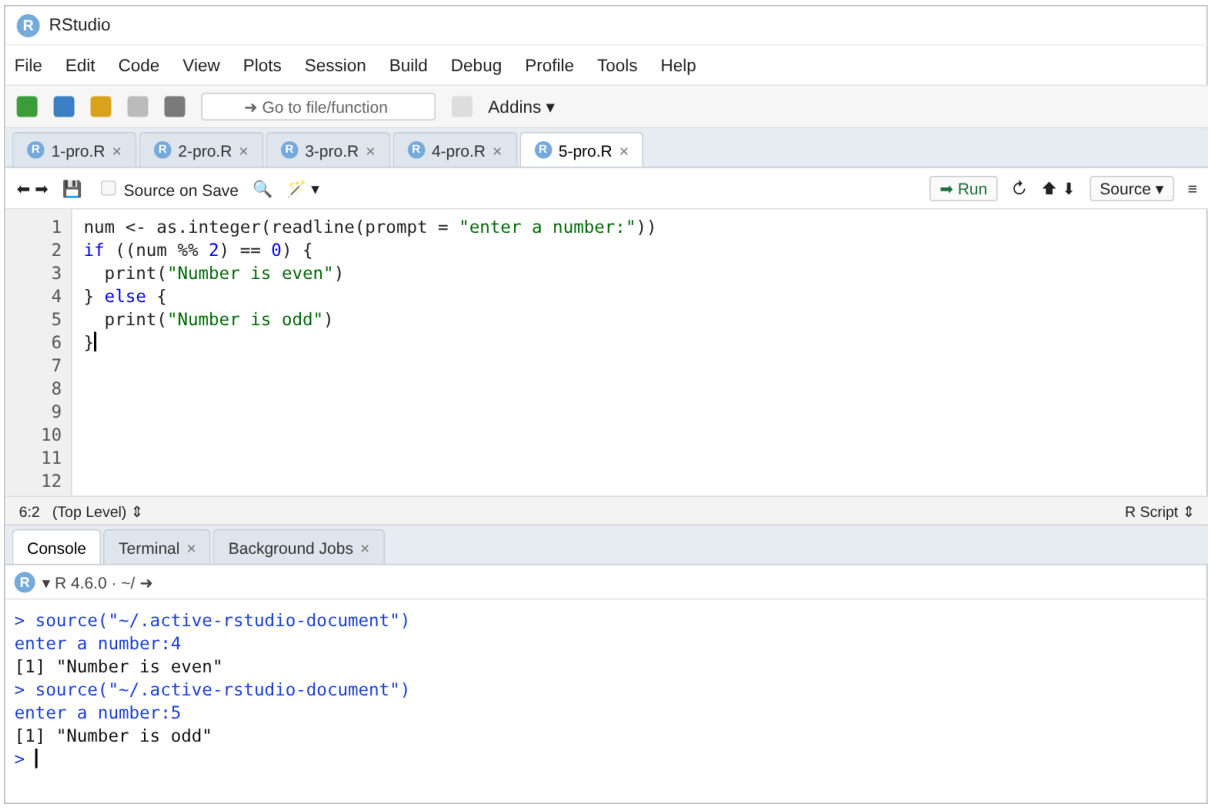
Coefficients:
  (Intercept)  BloodPressure        Glucose
       4.1421         0.1868         0.1617

[1] 4.14214
[1] 0.1867667
[1] 0.1617377
[1] 38.492
>
```

EXP-15


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
3-pro.R
4-pro.R
Source on Save
Run
Source
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 / num2
print(num3)
4:12 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
enter the first number:10
enter the second number:2
[1] 5
>

EXP-16


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
1-pro.R
2-pro.R
3-pro.R
4-pro.R
5-pro.R
Source on Save
Run
Source
num <- as.integer(readline(prompt = "enter a number:"))
if ((num %% 2) == 0) {
print("Number is even")
} else {
print("Number is odd")
}
6:2 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.0 · ~/
> source("~/.active-rstudio-document")
enter a number:4
[1] "Number is even"
> source("~/.active-rstudio-document")
enter a number:5
[1] "Number is odd"
>

EXP-17

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

→ Go to file/function | Addins ▾

1-pro.R × | 2-pro.R × | 3-pro.R × | 4-pro.R × | 5-pro.R × | 6-pro.R × | 7-pro.R ×

Source on Save | Run | Source ▾

```
1  names <- c("siri", "mahi", "chiru", "loki")
2  age <- c(23, 24, 24, 35)
3  marks <- c(88, 78, 25, 60)
4  df <- data.frame(names, age, marks)
5
6  print(summary(df$age))
7
8  write.csv(df, "datafr.csv")
9
10
11
12
```

8:28 (Top Level) ⇕ R Script ⇕

Console | Terminal × | Background Jobs ×

▾ R 4.6.0 · ~/ ➜

```
> source("~/.active-rstudio-document")
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  23.00   23.75   24.00   26.50   26.75   35.00
>
```

EXP-18

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

→ Go to file/function | Addins ▾

11-pro.R × | 12-pro.R × | 13-pro.R × | 14-pro.R ×

Source on Save | Run | Source ▾

```
1  a <- c(55, 67, 89, 80, 90)
2
3  par(mfrow = c(1, 2))
4  barplot(a, main = "Bar Plot", col = "skyblue")
5  barplot(a, horiz = TRUE, main = "Horizontal Bar Plot", col = "skyblue")
6
7
8
9
10
11
12
```

5:72 (Top Level) ⇕ R Script ⇕

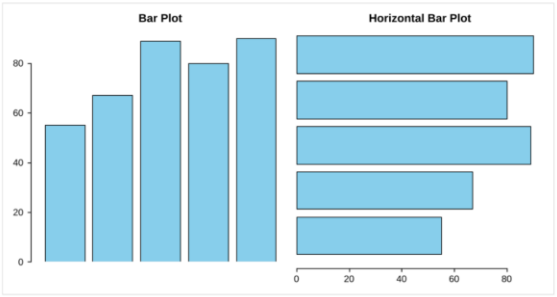
Console | Terminal × | Background Jobs ×

▾ R 4.6.0 · ~/ ➜

```
> source("~/.active-rstudio-document")
>
```

Files | Plots | Packages | Help | Viewer

Zoom | Export ▾



EXP-19

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

12-pro.R | 13-pro.R | 14-pro.R | 15-pro.R

```
1  names <- c("siri", "mahi", "chiru", "loki")
2  age <- c(23, 24, 24, 35)
3  marks <- c(88, 78, 25, 60)
4  df <- data.frame(names, age, marks)
5
6  boxplot(df$age, main = "Box Plot of Age", ylab = "Age", col = "lightblue")
7
8
9
10
11
12
```

6:75 (Top Level) R Script

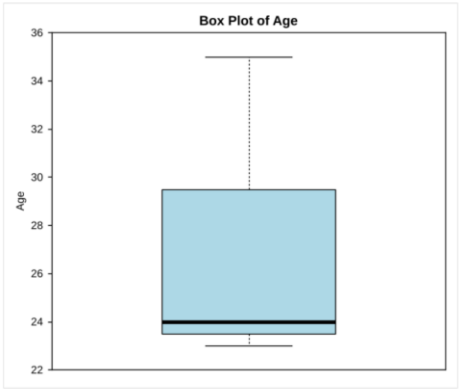
Console | Terminal | Background Jobs

R 4.6.0 · ~/

```
> source("~/.active-rstudio-document")
>
```

Files | Plots | Packages | Help | Viewer

Zoom Export



EXP-20

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

15-pro.R | 16-pro.R | 17-pro.R | 18-pro.R

```
1  diabetes <- data.frame(
2    Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
3    BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
4    Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
5  )
6
7  plot(diabetes$Age, diabetes$Glucose,
8       main = "Scatter Plot of Age vs Glucose",
9       xlab = "Age", ylab = "Glucose",
10      col = "blue", pch = 19)
11
12
```

10:29 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.6.0 · ~/

```
> source("~/.active-rstudio-document")
>
```

Files | Plots | Packages | Help | Viewer

Zoom Export

